

LabWork 1

Installation of MS SQL server 2012/2014. Data Bases files management. Managing SQL Server Security. Auditing Data Access and Encrypting Data

Part 1.

Prerequisites:

1. 2 VMs of Windows server 2012 R2 with GUI in Internal Network mode
2. At list 1 shared folder on one of machines
3. ISO image of Windows server is available in OS (to install Net.Framework 3.5 from it)
4. 1 additional Disk is attached to each server and ready to be operated

Tasks:

Install SQL server instances to both machines:

VM 1 should run default instance

VM 2 should run named instance

Properties for installations:

- All instances should have custom place of DB backups locations (not in the system drive)
- Leave service account in default configuration
- Mixed mode is selected for authentication (SQL + Windows)
- SQL Features to install:
 - SQL engine
 - Replication
 - SQL management tools – full installation
 - Others – by your choice, they aren't needed for this lab

Lab checklist:

1. There are SQL instances on the VMs. One is named, second is default. Read Windows feature installation menu – there is an option to use 'sources' folder from image
2. "Sa" SQL login and windows user can connect to SQL engine via SSMS locally on each VM. Execution of "select @@SERVERNAME" returns right instance names
3. At least "backups" folders are mapped to the location not in the system drive (Server properties – Database settings)
4. Student can connect from one machine to another using SQL login, and connect to both servers from Student host. RDP from Student host to VMs also available not via Hyper V tools. Windows firewall is active on both VMs.

Extra task:

Prerequisites:

1. PowerShell module SQLServer or SQLPS is installed on local student computer
2. Windows Firewall on VMs is active
3. SSMS is installed on local student computer

Task:

1. Set up connectivity from local PC to SQL instances using PowerShell and SSMS. Windows firewall is active. Checklist: Student can connect using PowerShell from local PS to both SQL instances (it is important: use only SQL login in NOT-AD area) and return "select @@ServerName" query result. Same from local SSMS.
2. Install SQL instance using modified configuration file, that was formed during first server installation (or create new one), (need screenshots of the process to confirm – 1 with console command that starts installation, the installation process, the console that finished command (no exceptions)). Install it as the 3rd SQL instance, to perform at least installation of named/default in case of fail.
3. Create all-in-one PS script, that:
 - installs Named instance,
 - set up firewall rules, that is needed for access to SQL and RDP,
 - Script returns name of installed SQL instance, and VM name in human readable view
 - Script returns list of installed SQL features in any human readable format
 - Script returns Firewall state and settings, that were changed by Student.

Part 2.

Prerequisites:

1. At least 1 running SQL server
2. At list 2 attached not system disk in the OS – to place users DBs files (ALL files of custom DBs)
3. Tools to work with SQL server

Pay attention to best practices and performance optimization for SQL server and databases.

Tasks:

1. Reconfigure TEMPDB to another database files location and options (use custom folder):

Main DB file:

Size: 10 MB

File growth: 5 MB

Maximum size: Unlimited

File location: on the attached disk

DB log file:

Size: 10 Mb

File growth: 1 MB

Maximum size: Unlimited

File location: on the attached disk

Check Several times what you are configuring. Read the course book for manual.

Restart SQL to apply changes

Delete old files

2. Create the **HumanResources** Database and the **InternetSales** Database:

Logical Name	Filegroup	Initial Size	Growth	Path
HumanResources	PRIMARY	50 MB	5MB/Unlimited	D:\Data\HumanResources.mdf
HumanResources_log		5 MB	1 MB/Unlimited	D:\Logs\HumanResources.ldf
InternetSales	PRIMARY	5 MB	1 MB / Unlimited	D:\Data\InternetSales.mdf
InternetSales_data1	SalesData	100 MB	10 MB / Unlimited	D:\Data\InternetSales_data1.ndf
InternetSales_data2	SalesData	100 MB	10 MB / Unlimited	D:\AdditionalData\InternetSales_data2.ndf
InternetSales_log		2 MB	10% / Unlimited	D:\Logs\InternetSales.ldf
Make the SalesData filegroup the default filegroup				

Extra task

1. Wrap DBs creation in PS Script, lab task 2 should be executed as non-stop script with Human readable output, all scripts start from Student local PC.

Options:

- Check if DBs with such names already exist, if yes – drop them with their files

- Check if everything DB created, files are in the correct places.
- Make Error handling and check, if it is working.

2. Using PowerShell, create and run script from local PC, which deploys:
Data Base “PCDRIVE” (with parameters like HumanResources)
DB contains tables with results of command:

`Get-PhysicalDisk | select -Property FriendlyName,BusType,HealthStatus,Size,MediaType`

Data types and names for columns could be selected by yourself

Compare pages and files sizes before and after filling data to DB (PS Script should show it in Human readable format).

Save logging of script actions with start and stop time of execution. (Decide what should be logged – it is your own choice).

Make error handling in the script.

Part 3.

1. Enable profiler session, that would show actions with database: queries and their results. Save file on disk. Choose the events, that you need for that, on your choice, basing on a description. Be ready to clarify why you choose them.
2. Create security audit, that would gather information about database access, logins to server and users' sessions. Make your choice with the same conditions as in previous step.
 - The purpose of the free choice of auditing is to simulate the situation when you the only engineer who knows security basics in the project, you can use only Google and documentation, and should perform the tasks, described above. Do not be scared, every answer in these steps is correct, but try to cover as much as possible security options according the task.

Write down chosen options, make screenshots of settings.

3. You need to set up security configuration for project team according next schema:

On first server (as DEV instance):	
5 developers:	Read and write the database data of User DB. No access to System DBs
1 application service (non-human user, service account)	Read/write/update data in the table of User DB. No access to System DBs
1 service account (non-application user, the service account for maintenance)	Modify user DB, create backups, but do not delete DB. Read systems DB.
1 user, who should make backups	Make backups of all DB, but cannot read data from User DB
2 QA users	Can only read data from user DB

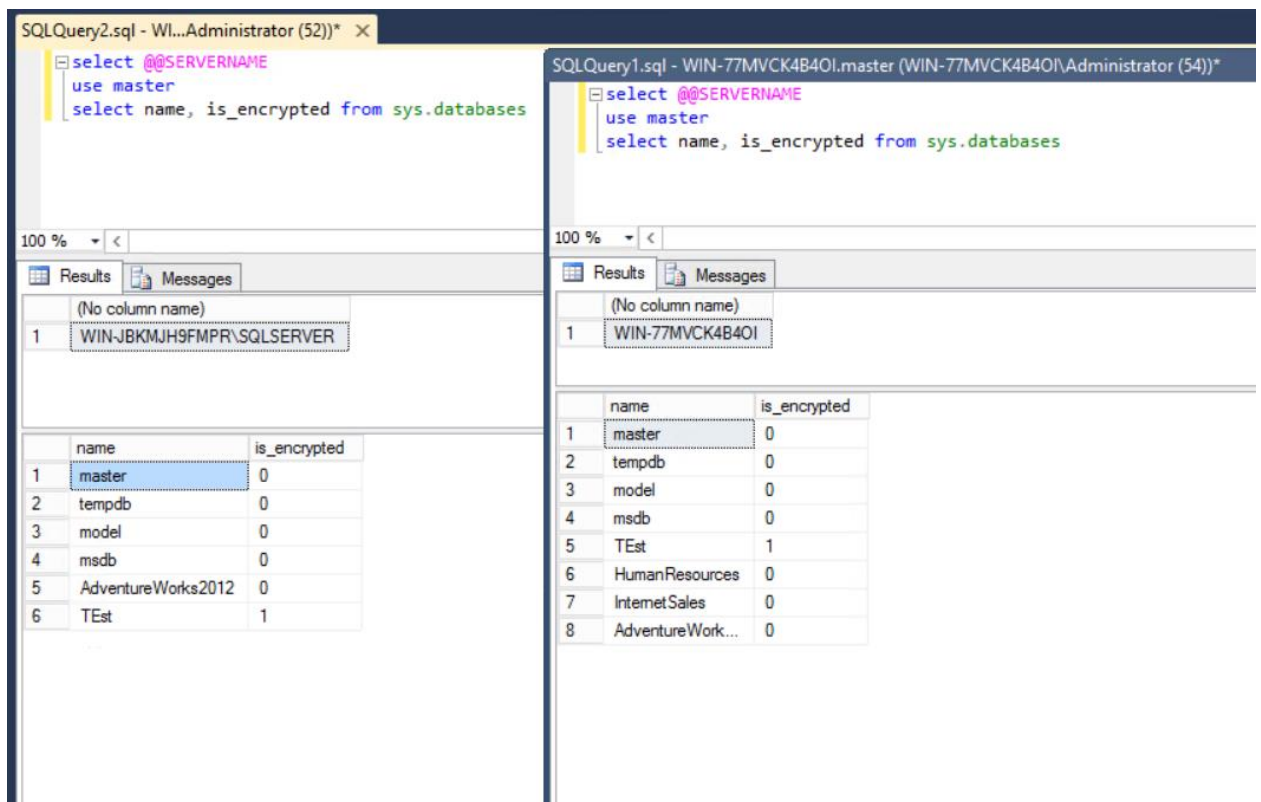
Make copy of Adventure works DB to User DB. Create Security configuration, defined in the task.

Check if each user has correct access. Make screenshots of the process.

Better to use Windows auth.

*You can create custom roles, but good practice is to do that only if you need very granular access, for example, data listener service should read ONLY one table from whole DB. So better to use basic roles.

4. Encrypt User database. Show that it is encrypted. Make compressed backup of encrypted DB. RESTORE (NOT JUST MOVE FILES) it on the second instance.
Screenshot to check should looks like:



5. Create security configuration on second instance according next plan:

On second server (as test instance):	
5 developers:	Read the database data of User DB. No access to System DBs
1 application service (non-human user, service account)	Read/write/update data in the table of User DB. No access to System DBs
1 service account (non-application user, the service account for maintenance)	Modify user DB, create backups, but do not delete DB. Read systems DB.
1 user, who should make backups	Make backups of all DB, but cannot read data from User DB
2 QA users	Can read data, write data, and update data in user DB

6. Review results of profiler. Find your requests. Export audit data to file. Filter it to show logins of all users, that was created in the task 3. Make screenshot for homework report and be ready to show log file later.

Extra Task

Make a PowerShell wrapper script to handle Users creation and database encryption (for one instance).

Extra for Extra task:

Read configuration with users from CSV file (columns: Function – DEV, test, service app, service user, backup user; Username – usernames; Password – User Passwords). May use the same config to create users in your AD/Local Computer accounts.